

# COMP 2121 Formula Sheet

| Rule of Inference                                                                                          | Related Logical Implication                                                                                 | Name of Rule                           |
|------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------|
| 1) $\frac{p \quad p \rightarrow q}{\therefore q}$                                                          | $[p \wedge (p \rightarrow q)] \rightarrow q$                                                                | Rule of Detachment<br>(Modus Ponens)   |
| 2) $\frac{p \rightarrow q \quad q \rightarrow r}{\therefore p \rightarrow r}$                              | $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$                                | Law of the Syllogism                   |
| 3) $\frac{p \rightarrow q \quad \neg q}{\therefore \neg p}$                                                | $[(p \rightarrow q) \wedge \neg q] \rightarrow \neg p$                                                      | Modus Tollens                          |
| 4) $\frac{p \quad q}{\therefore p \wedge q}$                                                               |                                                                                                             | Rule of Conjunction                    |
| 5) $\frac{p \vee q \quad \neg p}{\therefore q}$                                                            | $[(p \vee q) \wedge \neg p] \rightarrow q$                                                                  | Rule of Disjunctive<br>Syllogism       |
| 6) $\frac{\neg p \rightarrow F_0}{\therefore p}$                                                           | $(\neg p \rightarrow F_0) \rightarrow p$                                                                    | Rule of<br>Contradiction               |
| 7) $\frac{p \wedge q}{\therefore p}$                                                                       | $(p \wedge q) \rightarrow p$                                                                                | Rule of Conjunctive<br>Simplification  |
| 8) $\frac{p}{\therefore p \vee q}$                                                                         | $p \rightarrow p \vee q$                                                                                    | Rule of Disjunctive<br>Amplification   |
| 9) $\frac{p \wedge q \quad p \rightarrow (q \rightarrow r)}{\therefore r}$                                 | $[(p \wedge q) \wedge [p \rightarrow (q \rightarrow r)]] \rightarrow r$                                     | Rule of Conditional<br>Proof           |
| 10) $\frac{p \rightarrow r \quad q \rightarrow r}{\therefore (p \vee q) \rightarrow r}$                    | $[(p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow [(p \vee q) \rightarrow r]$                       | Rule for Proof<br>by Cases             |
| 11) $\frac{p \rightarrow q \quad r \rightarrow s \quad p \vee r}{\therefore q \vee s}$                     | $[(p \rightarrow q) \wedge (r \rightarrow s) \wedge (p \vee r)] \rightarrow (q \vee s)$                     | Rule of the<br>Constructive<br>Dilemma |
| 12) $\frac{p \rightarrow q \quad r \rightarrow s \quad \neg q \vee \neg s}{\therefore \neg p \vee \neg r}$ | $[(p \rightarrow q) \wedge (r \rightarrow s) \wedge (\neg q \vee \neg s)] \rightarrow (\neg p \vee \neg r)$ | Rule of the<br>Destructive<br>Dilemma  |

**No Name identity:**  $p \rightarrow q \Leftrightarrow \neg p \vee q$

**Contrapositive:**  $p \rightarrow q \Leftrightarrow \neg q \rightarrow \neg p$

Permutations/Combinations – without repetition:

$$P(n, r) = \frac{n!}{(n-r)!} \quad C(n, r) = \binom{n}{r} = \frac{n!}{(n-r)!r!}$$

Permutations/Combinations – with repetition:

$$P(n, r) = n^r \quad C(n+r-1, r) = \binom{n+r-1}{r}$$

Given  $n$  items with  $n_1$  alike,  $n_2$  alike, ...,  $n_k$  alike, the number of permutations of all  $n$  items is:

$$\frac{n!}{n_1!n_2!\cdots n_k!}$$

Set Theory:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

### The Laws of Logic

For any primitive statements  $p, q, r$ , any tautology  $T_0$ , and any contradiction  $F_0$ ,

$$1) \neg\neg p \Leftrightarrow p \quad \text{Law of Double Negation}$$

$$2) \neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q \quad \text{DeMorgan's Laws}$$
$$\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$$

$$3) p \vee q \Leftrightarrow q \vee p \quad \text{Commutative Laws}$$
$$p \wedge q \Leftrightarrow q \wedge p$$

$$4) p \vee (q \vee r) \Leftrightarrow (p \vee q) \vee r^\dagger \quad \text{Associative Laws}$$
$$p \wedge (q \wedge r) \Leftrightarrow (p \wedge q) \wedge r$$

$$5) p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r) \quad \text{Distributive Laws}$$
$$p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$$

$$6) p \vee p \Leftrightarrow p \quad \text{Idempotent Laws}$$
$$p \wedge p \Leftrightarrow p$$

$$7) p \vee F_0 \Leftrightarrow p \quad \text{Identity Laws}$$
$$p \wedge T_0 \Leftrightarrow p$$

$$8) p \vee \neg p \Leftrightarrow T_0 \quad \text{Inverse Laws}$$
$$p \wedge \neg p \Leftrightarrow F_0$$

$$9) p \vee T_0 \Leftrightarrow T_0 \quad \text{Domination Laws}$$
$$p \wedge F_0 \Leftrightarrow F_0$$

$$10) p \vee (p \wedge q) \Leftrightarrow p \quad \text{Absorption Laws}$$
$$p \wedge (p \vee q) \Leftrightarrow p$$